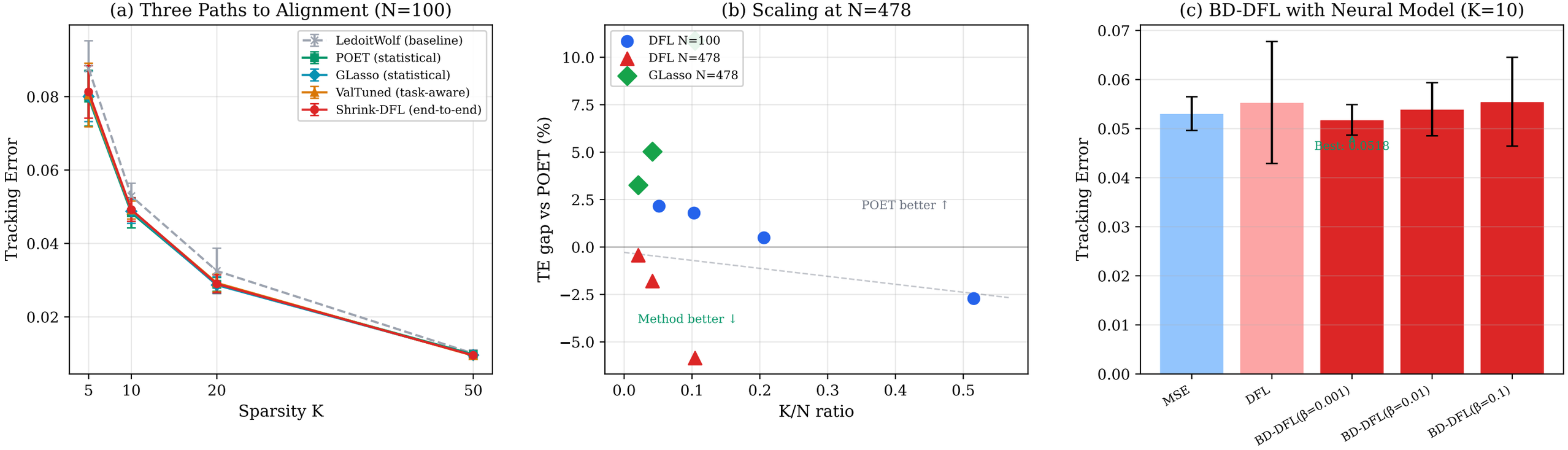
(a) Three Paths to Alignment (N=100)
LedoitWolf (baseline)
POET (statistical)
GLasso (statistical)
ValTuned (task-aware)
Shrink-DFL (end-to-end)
Tracking Error
0.08
0.06
0.04
0.02
5
10
20
50
Sparsity K
(b) Scaling at N=478
DFL N=100
DFL N=478
GLasso N=478
TE gap vs POET (%)
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
POET better ↑
Method better ↓
0.0
0.1
0.2
0.3
0.4
0.5
K/N ratio
(c) BD-DFL with Neural Model (K=10)
Tracking Error
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
Best: 0.0518
MSE
DFL
BD-DFL(β=0.001)
BD-DFL(β=0.01)
BD-DFL(β=0.1)